

COMP 271 Diagnostic Check (from COMP 170)

A quick check of COMP 170 fundamentals before Week 01. No grade attached — just a way to see where things stand.

1. Variables & Mutability

What will be printed?

```
a = [1, 2, 3]
b = a
b.append(4)
print(a)
```

- (A) [1, 2, 3]
- (B) [1, 2, 3, 4]
- (C) [4]
- (D) Error

2. Function Calls & Assignment

What is printed?

```
def mystery(x):
    x = x + 1
    return x

y = 5
mystery(y)
print(y)
```

- (A) 6
- (B) 5
- (C) None
- (D) Error

3. Lists vs Tuples

Which statement is true?

- (A) Lists are immutable, tuples are mutable
- (B) Tuples allow item reassignment
- (C) Lists allow item reassignment
- (D) Lists and tuples behave identically

4. Loop Execution

How many times does the loop run?

```
for i in range(2, 10, 2):
    print(i)
```

- (A) 3
- (B) 4
- (C) 5
- (D) 6

5. Conditionals

What is printed?

```
x = 7
if x % 2 == 0:
    print("Even")
elif x > 5:
    print("Big")
else:
    print("Odd")
```

- (A) Even
- (B) Odd
- (C) Big
- (D) Nothing

6. List Indexing

What does this print?

```
lst = [10, 20, 30, 40]
print(lst[-1])
```

- (A) 10
- (B) 20
- (C) 30
- (D) 40

7. Boolean Logic

What is the value of the expression?

```
True and not False or False
```

- (A) True
- (B) False
- (C) Error
- (D) Depends on parentheses

8. Dictionaries

Which operation retrieves a value from dictionary d using key "id"?

- (A) d("id")
- (B) d[id]
- (C) d["id"]
- (D) d.get[id]

9. Classes & Objects

What does self represent in a Python class?

- (A) The class itself
- (B) A static variable
- (C) The current object instance
- (D) A global reference

10. Method Calls

Given the class below, what does this code print?

```
class Counter:
    def __init__(self):
        self.value = 0

    def inc(self):
        self.value += 1

c = Counter()
c.inc()
print(c.value)
```

- (A) 0
- (B) 1
- (C) Error
- (D) None

11. Variable Scope

What is printed?

```
x = 10

def f():
    x = 5

f()
print(x)
```

- (A) 5
- (B) 10
- (C) Error
- (D) None

12. List Operations

Which operation adds an element to the end of a list?

- (A) lst.add(x)
- (B) lst.insert(x)
- (C) lst.append(x)
- (D) lst.push(x)

13. Return vs Print

Which statement is correct?

- (A) print sends a value back to the caller
- (B) return displays output on screen
- (C) return sends a value back to the caller
- (D) print ends function execution

14. Performance Intuition (Pre-DS)

Which operation is generally fastest for large lists?

- (A) Searching for an element
- (B) Appending an element
- (C) Removing from the middle
- (D) Sorting the list

15. Reading Code

What does this function compute?

```
def f(lst):  
    total = 0  
    for x in lst:  
        total += x  
    return total
```

- (A) Maximum value
- (B) Average
- (C) Sum of elements
- (D) Number of elements